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Ground Handling Loads

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Introduction

Although they're often in the shadow of flight and parachute deployment loads, ground handling loads can be important. Ground handling failures, while they can usually be repaired, often still have important schedule impacts.

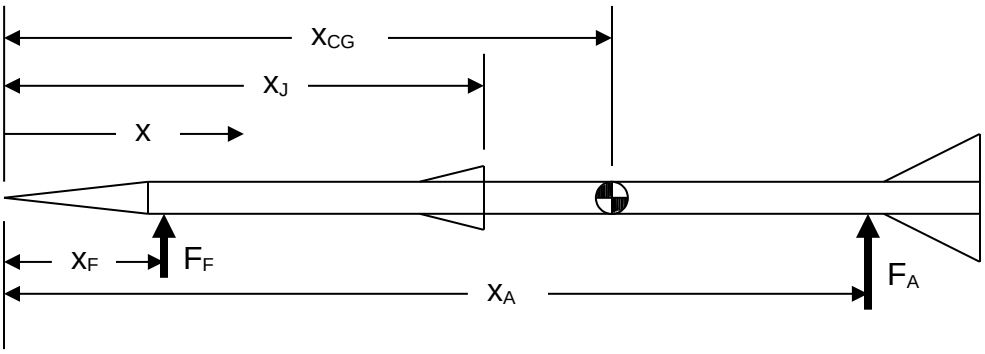
There are two known ground handling loading conditions: The first is associated with bending loads on a slender rocket carried from each end while transported. The second is tearing extraction of riding lug/button while the rocket is being loaded on the launcher rail.

Nomenclature

Mnemonic	Definition
x	Axial coordinate,
x_J	Axial location of a joint,
x_{CG}	Axial location of the center of mass,
x_F	Axial location of the forward support force,
x_A	Axial location of the aft support force,
x_L	Axial location of the aft riding lug/button,
F_F	Forward support force,
F_A	Aft support force,
F_L	Force on the aft riding lug, positive in tension,
W	Rocket weight,
K	Safety factor
N	Load factor,
w_i	W eight of the ith body element,
x_i	Axial location of the center of mass of the ith body element, and
M	Bending moment,
S	Aft button shear force

Transportation Loads

Consider a slender rocket carried at each end while transported. Assume there's a critical potential weakness at some midpoint, e.g., an interstage joint. See the sketch below:



The equations of statics are

$$F_F + F_A = W \text{ , and}$$
$$F_F (x_{CG} - x_F) = F_A (x_A - x_{CG}) . \tag{1}$$

These may be solved for the two support forces:

$$F_F = W \frac{x_A - x_{CG}}{x_A - x_F} \text{, and}$$
$$F_A = W \frac{x_{CG} - x_F}{x_A - x_F} \tag{2}$$

Next, decompose the body into a sequence of elements with the weight and center of mass of each being w_i and x_i. Suppose further that the body were subject to a phantom "cut" at station x_J, assumed to be aft of x_F. Consider the front piece. A shear force and bending moment must act at the phantom cut to maintain equilibrium. The shear force is just the difference between the weight forward of x_J and the forward support force. Similarly, the bending moment is

$$M = F_F x_F - \sum_1^J w_i x_i$$

But, this is not the total story. While being moved, the rocket will be subjected to bouncing and jouncing. We account for this by using a load factor:

$$M = N(F_F x_F - \sum_1^J w_i x_i) \tag{3}$$

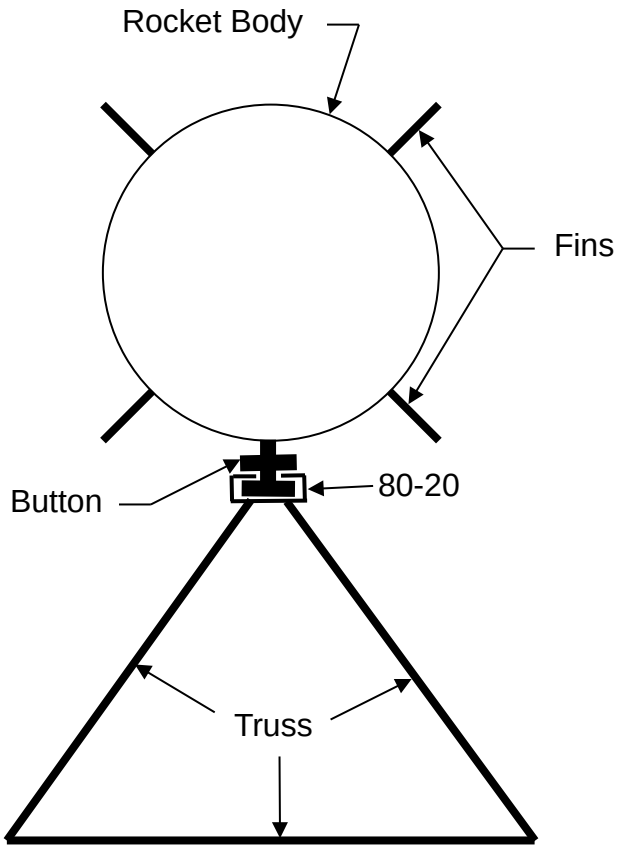
What value should be used for N? All designers will be must happier using a value too high (engendering a heavier structure) rather then a value slightly too small. I suggest that we take N = 2.

Finally note that these results can be readily implemented in BENDIT, the Excel software for loads estimation.

Riding Button Extraction

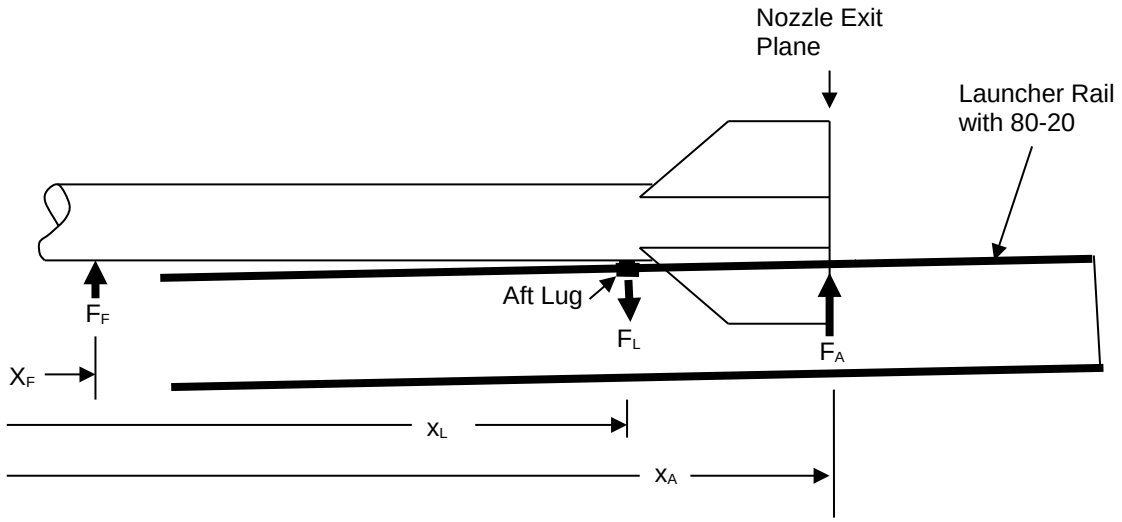
The riding lug/button failure has been observed at least twice. The relevant operations concept uses a rail of 80-20™ aluminum. 80-20™ is a slotted precision extrusion designed to be used as soft tooling for assembly and manufacturing. When used as part of a launcher, cylindrical plastic buttons attached to a rocket ride in the 80-20™ slot to initially guide the rocket's flight. The 80-20™ is attached to the top side of a rail truss.

To install the rocket on the rail, the rail is first lowered to a horizontal position where ground crew holds it in place. A second ground crew slides the aftermost (of 2) rocket lug into the free end of the 80-20™ slot. The problem comes as the second crew moves the rocket down the rail and tries to engage the forward button. This often results in manhandling the aft lug to the point where its attachment to the rocket fails.



The notional sketch above shows the general arrangement. Inserting the forward button (often near the rocket midpoint) is difficult because it is not easily seen from above. The natural tendency is to elevate the nose to get a better view. Direct calculation with typical dimensions shows that a pitch-up of only 1.5^{o} to 5°, or less is sufficient to bring

the nozzle exit plane into contact with the rail. If the button attachment has a small compliance (not much stiffness) it is possible to go beyond the 5°, or so without significant kinesthetic feedback. It works just like a claw hammer extracting an undesired nail, only this time it's the lug/button that's extracted.



What's the remedy? Next, look at the character of the nose force, F_F, generated by a ground crew member. Initially, before the rocket body exit plane comes into contact with the rail, it is that given by eq.(2) in the previous section:

$$\frac{F_F}{W} = \frac{x_L - x_{CG}}{x_L - x_F} \tag{4}$$

Here, the lug force F_L is supporting the aft end of the rocket by pushing up. Now, following eq's. (1), we can write the equilibrium equation for the moments about the CG:

$$F_F(x_{CG} - x_F) + F_L(x_L - x_{CG}) = F_A(x_A - x_{CG}),$$

together with the equilibrium equation for the forces:

$$F_A + F_F = W + F_L$$

Eliminating F_A results in

$$F_F(x_{CG} - x_F) + F_L(x_L - x_{CG}) = (W + F_L - F_F)(x_A - x_{CG}), \text{ or}$$

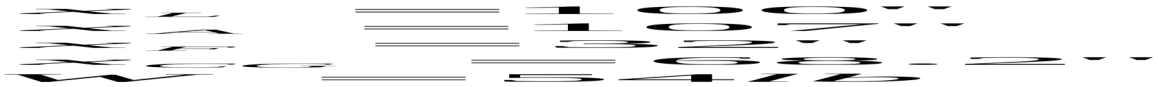
$$F_F(x_A - x_F) + F_L(x_L - x_A) = W(x_A - x_{CG}) \tag{5}$$

Now, the angular change as F_F increases is too small to make an effective feedback mechanism. But the increase in force, if large enough, can be used as a warning signal to a ground crewman. Suppose the warning threshold were K times the initial level given by eq. (4). Then, if the lug/button were designed to a limit load of

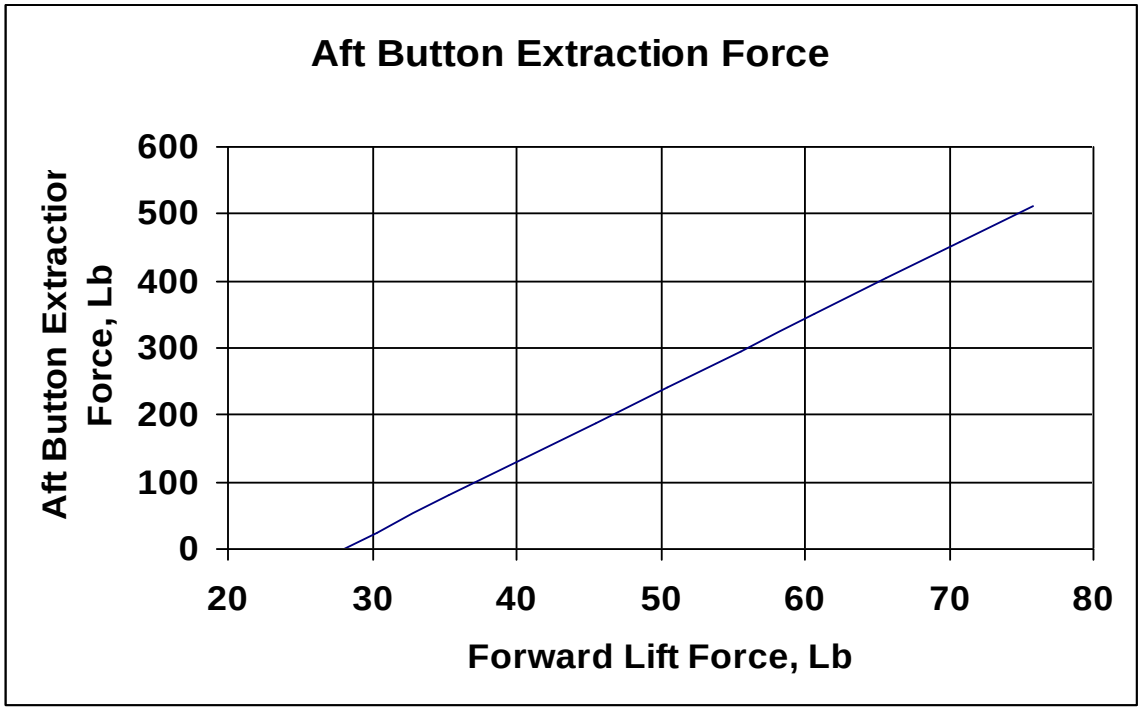
$$F_L = W \frac{K(x_L - x_{CG})(x_A - x_F) - (x_A - x_{CG})(x_L - x_F)}{(x_L - x_F)(x_A - x_L)}, \qquad (6)$$

then the button would not be extracted

Now, consider an ESRA typical example. Take



Then, eq.(5) has been coded into an Excel file named UNBUTTON.xls, and the results shown below. Here the lift force is that applied by a ground crewman pushing up near the nose, and the button extraction force is the tension load attempting to pull the button off the rocket.



Absent better knowledge, it's recommended that K = 2 be used. . In this example, the limit extraction load would be about 243 lb. corresponding to a lift force of 51 lb.(K = 2). Yes, I have known ground crewmen who did not know their own strength. Do not assign such individuals to the delicate task of lifting the rocket nose.

Finally, it's recommended that both forward and aft lugs be designed to the same criteria. Also note that if the buttons are attached with round head fasteners, the top of the heads may rub against the bottom of the 80-20™ channel. Therefore, use countersunk fastener heads.

Aft Riding Button Shear

Beside the extraction load estimated above, the aft riding button must support the entire weight of the rocket when it is erect on the rail. Absent better insight, it is suggested that the same safety factor used to develop the button extraction load be used for the button shear force:

$$S = KW .$$